

TASK 2: VIRTUAL LAB SETUP & PRACTICAL CONFIGURATION

Internship: Ethical Hacking Internship

Submitted By: Mogiipuram Tejaswini

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Task 2 Report: Practical Laboratory Virtualization and Setup

1. Introduction & Task Alignment

In today's digital landscape, modern applications and storage services deal with immense volumes of sensitive data, rendering robust protection systems mandatory. As outlined in the fundamental theory of security models, cybersecurity is defined as the foundational practice of safeguarding computer configurations, network domains, and data repositories from unauthorized malicious interference.

While the underlying objective remains the preservation of information integrity, confidentiality, and availability, defensive elements are significantly enhanced via proactive vulnerability discovery. This is the domain of Ethical Hacking: a legal framework wherein specialized tools and common methodologies are applied under controlled configurations to systematically uncover flaws before malicious adversaries exploit them.

Legal Reminder & Ethical Mandate: Always obtain formal, written permission before testing any system or computer architecture. Ethical hacking operations focus completely on defensive security posture enhancement and must adhere strictly to predefined authorization boundaries.

2. Core Systems Pre-requisite & Validation Check

Before initiating the architecture setup, a comprehensive hardware baseline check was executed on the underlying host operating system. This verification step confirms that hardware-level abstraction extensions are active, which ensures that complex, 64-bit guest environments operate efficiently without causing disruption or performance degradation to production systems.

- **Host Processor Infrastructure:** Intel(R) Core(TM) i7-9750H CPU running at a baseline clock rate of 2.60GHz, utilizing 6 hardware cores and 12 logical processing paths.
- **System Memory Capacity:** 32.0 GB installed RAM, providing ample capacity for high-performance guest instance scaling.

- **Hardware Virtualization State:** Verified as 'Enabled' directly within the core BIOS/UEFI firmware layer.

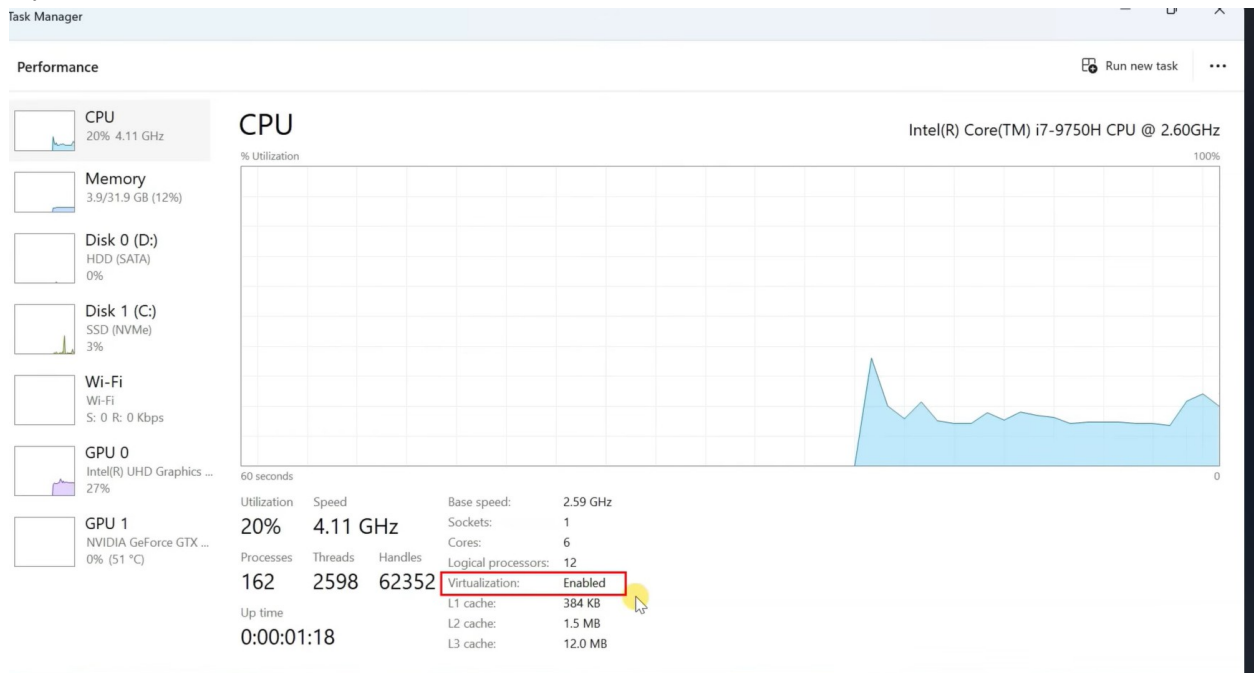


Figure: Visual verification of windows task manager - hardware virtualization status check.

3. Step-by-Step Hypervisor and Environment Installation

Step 01: Deploying the Hosted Virtualization Engine

To construct an isolated, safe space for learning and technical experimentation, the Oracle VirtualBox Setup Wizard (Version 7.1.6) was initiated. Standard setup components, including default virtual core interfaces and network management modules, were configured using standard settings.

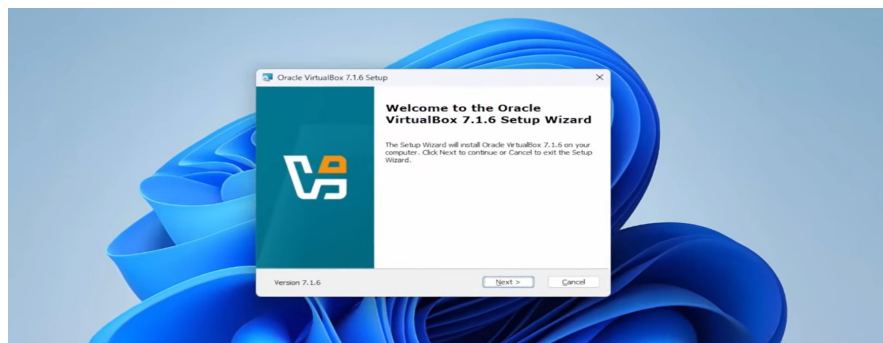


Figure: Visual verification of oracle virtualbox 7.1.6 setup wizard - welcome screen.

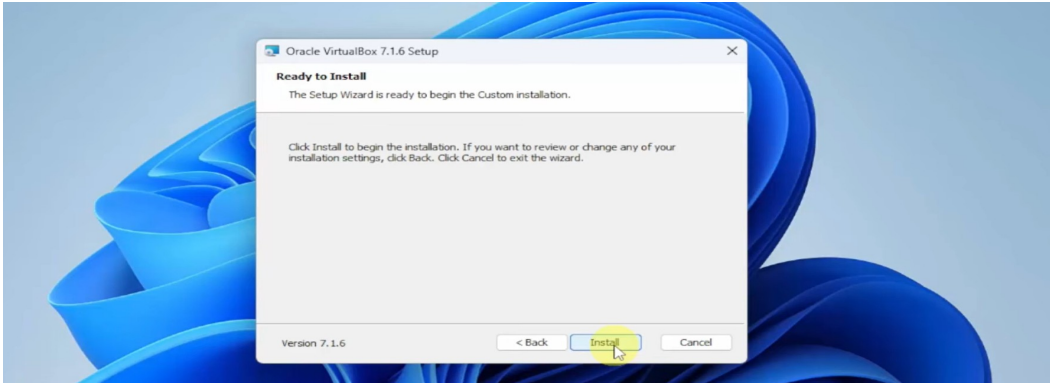


Figure: Visual verification of oracle virtualbox 7.1.6 custom setup - ready to install.

Step 02: Creating and Customizing the Attacker Virtual Station

Using the primary management console, a new virtual machine profile was constructed. To map resources appropriately according to the security workflow requirements, the system properties were defined manually via the advanced operational wizard:

- **Target Workstation Name:** Kali Linux
- **Operating System Subtype:** Debian (64-bit) baseline kernel layout definition.
- **System RAM Allocation:** 8192 MB (8 GB) assigned to support high-performance tool executions.
- **Processor Cores:** 6 vCPUs selected from the multi-core allocation window.
- **Storage Allocation Scheme:** Virtual Disk Image (VDI) configuration utilizing a dynamically allocated virtual hard disk setup strategy.

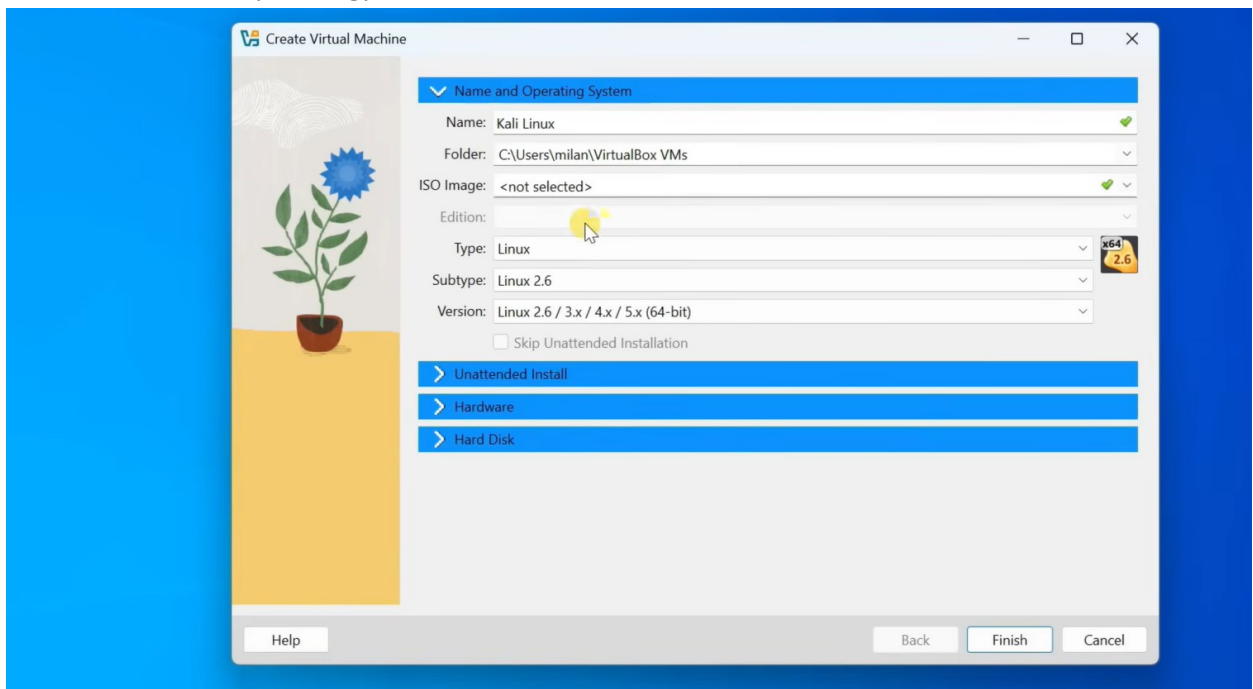


Figure: Visual verification of oracle virtualbox manager - creating the virtual machine.

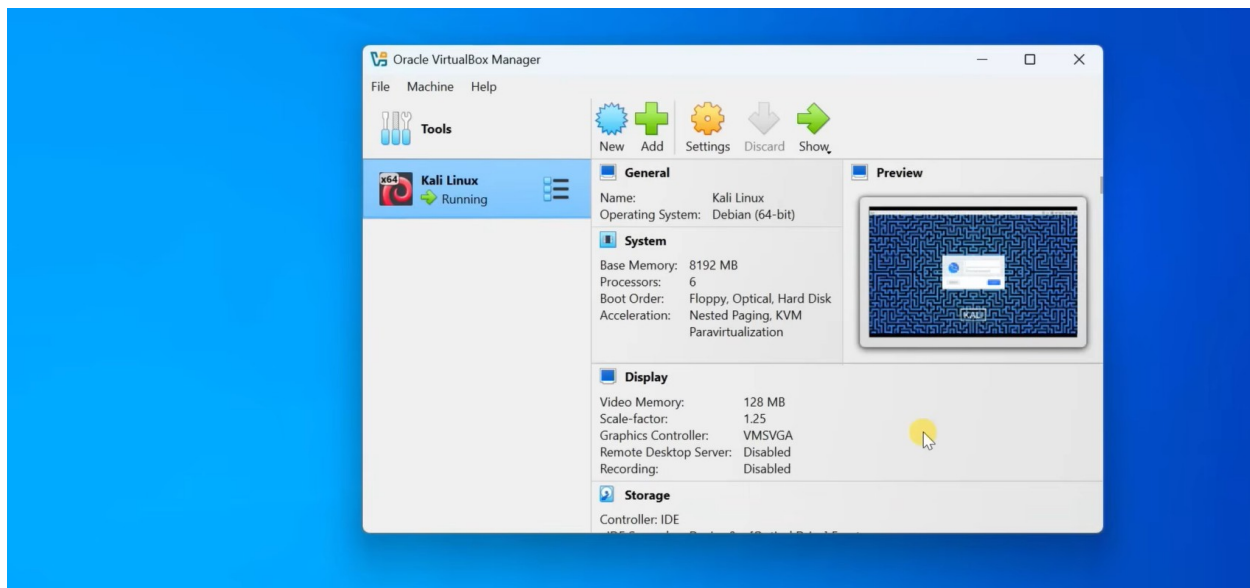


Figure: Visual verification of oracle virtualbox manager - virtual hardware configuration workspace.

Step 03: Advanced Core Infrastructure & Network Settings

To protect real-world systems, network and storage isolation layers must be configured explicitly within the advanced virtual machine settings panel before firing up the guest OS:

- **Hardware-Assisted Acceleration:** Nested Paging and KVM Paravirtualization interfaces enabled to achieve efficient execution rings.
- **Isolated Network Topology:** The Network Adapter was bound to Network Address Translation (NAT) mode to keep internal testing traffic contained while allowing safe web access for package maintenance.

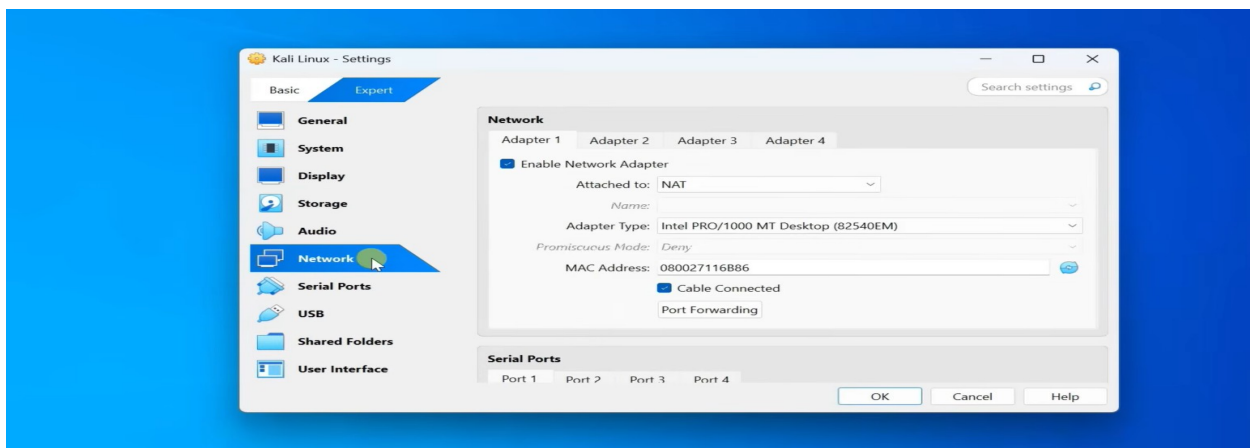


Figure: Visual verification of virtual machine advanced settings - network adapter configuration panel.

4. Operating System Installation and Workspace Initialization

The virtual workstation profile was successfully booted by mapping the official Kali Linux installer image to the virtual optical media controller. This initiated the standard installer sequence, requiring configuration of the core network environment properties, host system labels, and individual administrative user accounts.

- **Target Host Name Attribute:** Configured explicitly with the system label 'kali'.
- **Primary User Creation:** A non-root system administrative account was generated using the username 'milan' along with corresponding secure password parameters.

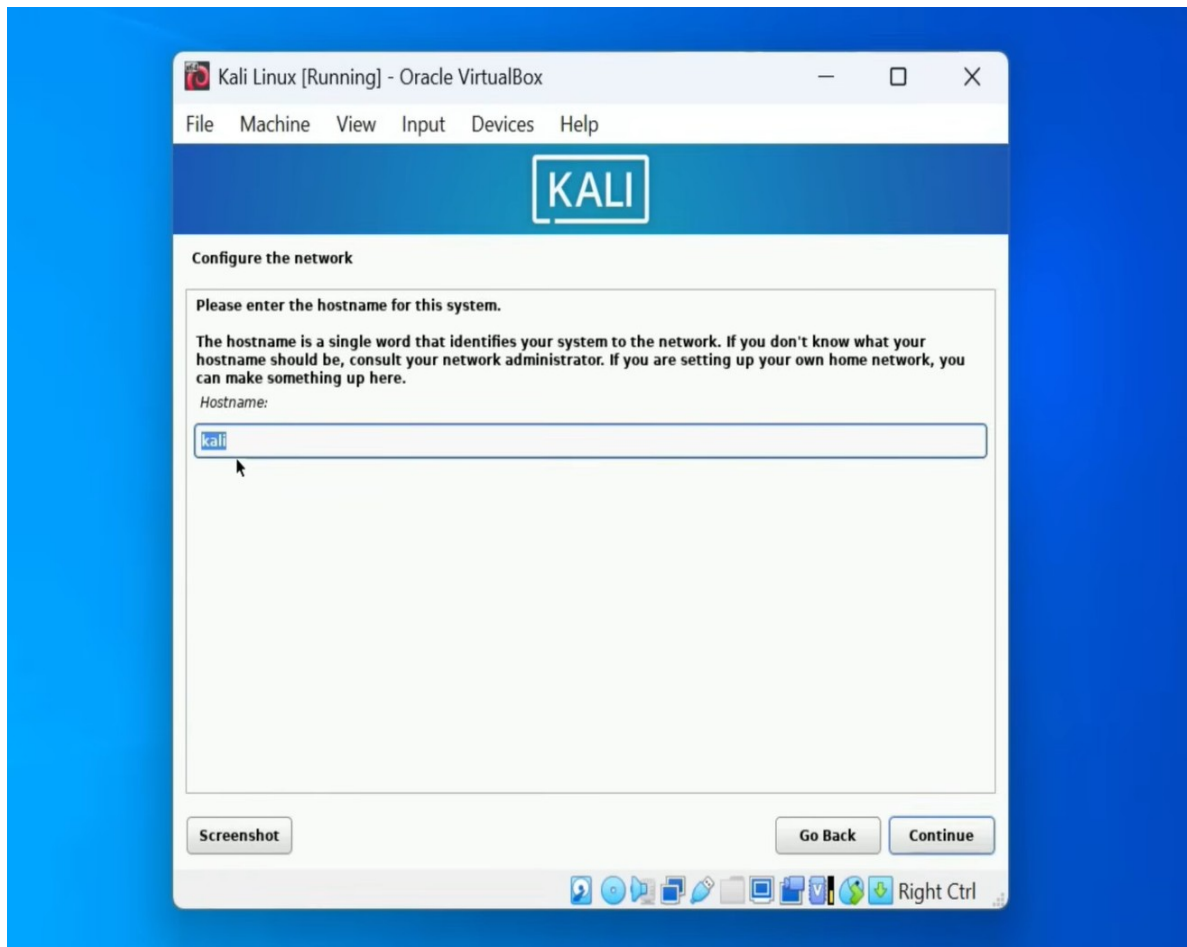


Figure: Visual verification of kali linux network configuration - setting the hostname.

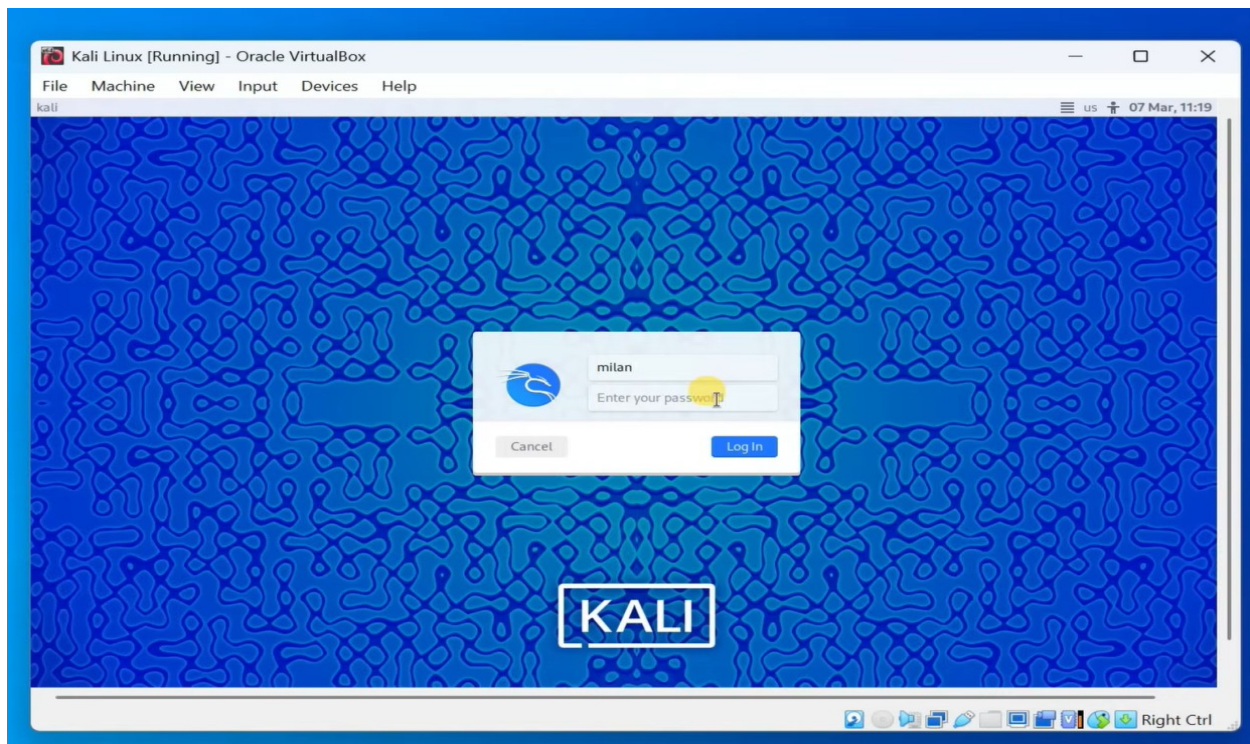


Figure: Visual verification of kali linux display manager - authentication interface for account 'milan'.

Following automated post-installation configuration runs, the system successfully loaded into the operational graphical XFCE user workspace desktop.

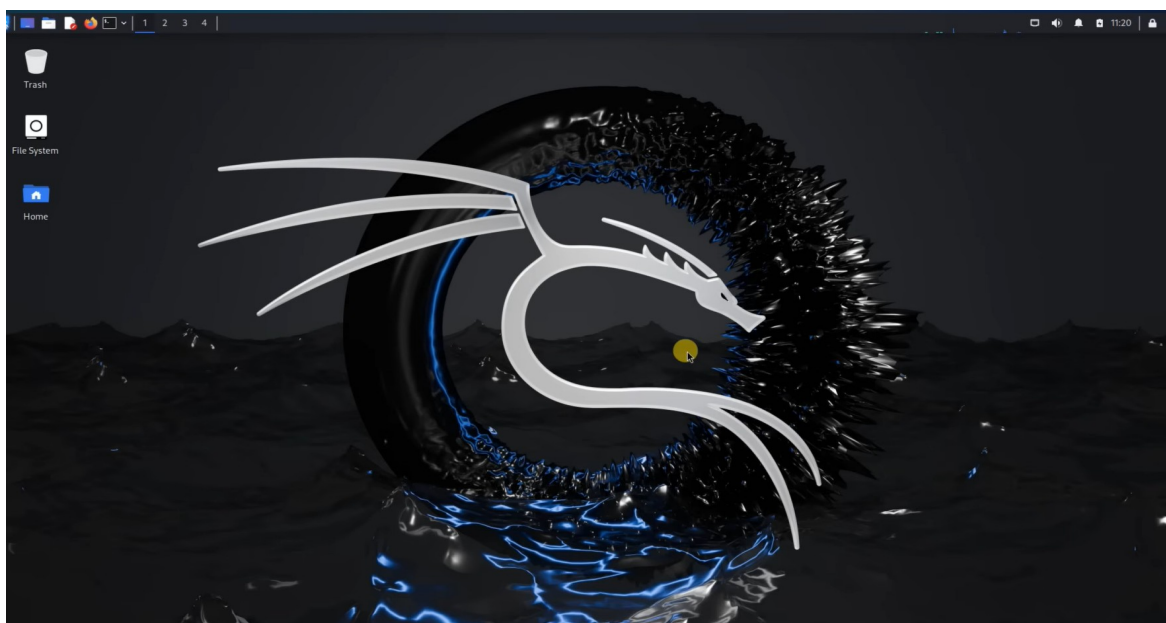


Figure: Visual verification of kali linux desktop environment - initial successful boot interface.

5. Laboratory Architecture and Target Node Integration

To fully support structured ethical hacking methodologies, the single host environment is expanded into a complete multi-tier laboratory topology. This ensures safe execution spaces for both offensive tooling operations and forensic tracking practice tasks:

Lab Environment Element	Technical Role & Purpose	Network Isolation Strategy
Kali Linux Instance	Attacker Station & Core Vulnerability Pentesting Toolset	NAT / Private Host-Only Virtual Switch
Windows VM	Target Workspace / Enterprise Endpoint Client Simulation	Internal Isolated Testing Subnet Only
DVWA / Metasploitable	Intentionally Vulnerable Lab Server Node for Flaw Analysis	Strictly Isolated Host-Only Domain Context

6. Analytical Conclusion & Validation Results

The practical realization of the virtual practice laboratory has been successfully completed and validated. By running an active instance inside an isolated hosted hypervisor, the testing setup meets strict information security principles. This platform is now fully optimized and prepared for executing advanced exploitation tests, checking network logging behavior, and building complex cybersecurity engineering implementations safely.